HyperCLIP-CD: Hyperbolic Multimodal Representation Enhancement for Remote Sensing Change Detection

Qian Yang, Shujun Zhang, Xiao Li, YuNuo Wang, and Yuanfei Zhou

***Abstract*—Remote sensing image change detection (RSCD) is an important branch of remote sensing image analysis, aimed at identifying changes in land features from bitemporal images. In recent years, change detection methods have predominantly been based on deep learning (DL) approaches, some even integrating foundational vision–language models to provide multimodal data, achieving impressive results. However, existing multimodal methods merely utilize multimodal data superficially, without delving into the representation relationship between vision and language. Recent research has proposed that non-Euclidean spaces, such as hyperbolic space, offer advantages in modeling graph–text relationships. To this end, we introduce HyperCLIP-CD, a multimodal network specifically designed for the RSCD task. It leverages the property of hyperbolic space, which is suitable for embedding tree-like data and combines CLIP to capture the hierarchical structure and non-Euclidean features of image and text data. Specifically, we have designed a hierarchical semantic textual prompting (HTP) generation method tailored for change regions that meet the requirements. Based on a dual-branch CLIP architecture, we extract multimodal features from bitemporal images and propose a hyperbolic multimodal semantic alignment (HMSA) module to obtain deeper semantic change information. HyperCLIP-CD has achieved higher detection accuracy than other advanced networks on both the binary change detection (BCD) datasets LEVIR-CD and CDD, as well as the semantic change detection (SCD) dataset SECOND.**

***Index Terms*—Change detection, deep learning (DL), hyperbolic geometry, multimodal artificial intelligence.**

I. Introduction

THE purpose of remote sensing image change detection (RSCD) is to identify semantic changes between a pair of remote sensing images of the same area taken at different times. Depending on the different requirements for change categories, RSCD tasks can be classified into binary change detection (BCD) and semantic change detection (SCD) [1]. RSCD has a wide range of applications in various fields such as land use, surface change, building and forest cover change, urban planning, disaster management, landslide detection, and many others [2], [3], [4], [5], [6]. Unlike natural images, remote sensing images possess unique characteristics such as high resolution, multispectral, and special viewing angles; these bring both advantages and new challenges to related image analysis. How to fully utilize the rich information contained in very-high-resolution (VHR) remote sensing images and the unique bi-temporal change characteristics of change detection tasks, while ensuring robustness in the results, has always been an important issue in related fields.

In recent years, deep learning (DL) methods have undergone rapid development and have gradually become mainstream in the field of RSCD. Initially, due to limitations in models and computational power, RSCD tasks mostly relied on pixel-based [7], [8] and object-based [9], [10], [11] methods. After the introduction of fully convolutional networks (FCNs) [12] in this field, the powerful feature extraction capabilities of convolutional neural networks (CNNs) made scene-based CD methods mainstream. In recent years, Siamese networks [13] have become the most popular baseline model due to their compatibility with the dual input characteristic of RSCD tasks. Many excellent RSCD methods based on the Siamese structure have emerged, encompassing not only CNN-based approaches [14], [15], [16] and Transformer architectures [17], [18], [19], but also the recently emerging Mamba-based methods that have gained increasing popularity [20], [21]. However, these methods are limited to single-modal data. As a special type of semantic segmentation task, it is necessary to introduce multimodal semantic information into RSCD tasks.

Recently, vision–language representation learning has gained popularity, with the characteristic of learning representations from images and their corresponding text promoting the development of many multimodal tasks, including cross-modal retrieval [22], large language models [23], and others. Among them, the contrastive language–image pretraining (CLIP) model [24] can process inputs from different modalities and generate comparable embeddings, thereby establishing a connection between text and images. Wang et al. [25] were among the first to introduce multimodal representation learning into remote sensing image interpretation by converting image metadata into text prompts and integrating them with image features through cross-modal fusion. In the field of RSCD, Dong et al. [26] pioneered the use of the CLIP pretrained model to generate text descriptions for CD datasets, enhancing image semantics by incorporating multimodal data, and proposed a multimodal CD model, ChangeCLIP.

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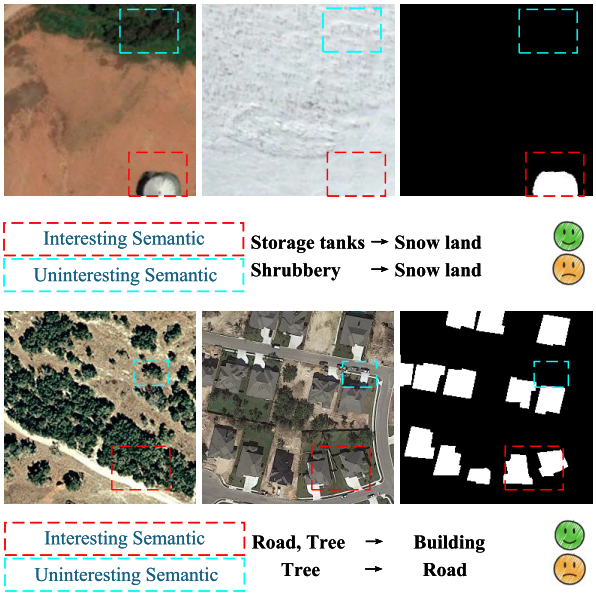


Fig. 1. Interested semantics in different CD datasets.

However, ChangeCLIP focused on introducing vision–text representation methods into CD tasks without deeper exploitation of multimodal information. Like most vision–text representation models [24], [27], ChangeCLIP represent images and text as vectors in a high-dimensional Euclidean affine space and normalize the embeddings to unit L2-norm. However, Euclidean space applies the same distance metric to all embedding points, making it difficult for such a geometric choice to capture the visual–semantic hierarchy. Research by Desai et al. [28] has shown that text and images, as well as text and text, exhibit conceptual hierarchies when represented as embedding vectors, and hyperbolic space can naturally handle such hierarchies [29]. Similarly, research by Bronstein et al. [30] has indicated that remote sensing images possess highly non-Euclidean latent characteristics, necessitating the introduction of hyperbolic properties into RSCD tasks. Furthermore, existing methods do not specifically explore the unique semantic requirements of CD tasks. As shown in Fig. 1, there is no uniform standard for semantic changes in CD tasks. Depending on the dataset, our interest in "changes" varies. In building change datasets, we often only focus on changes in buildings and ignore other semantics, even if they fall within the scope of "potential changes." In contrast, multiclass SCD focuses on more than five types of semantic changes, but this still does not encompass all visible changes to the human eye.

In this article, we design a hierarchical semantic textual prompting (HTP) generation method that categorizes cues based on data requirements. Leveraging the powerful image recognition and zero-shot classification capabilities of the basic CLIP model, we create specific classification cues for different datasets based on 56 common land-cover types [26], taking into account the different labeled areas of interest in each dataset. We categorize these cues under the root nodes of "semantic categories of interest (SCoI)" and "semantic categories of no interest (SCoNI)," hierarchically encompassing the semantic classes of the dataset. Based on the generated text cues, we construct a change detection network called HyperCLIP-CD, which integrates vision–text representation learning. In the encoding stage, we combine the Siamese network structure with the image and text encoders of CLIP to extract features from images and text. These features are then mapped from Euclidean multimodal space to hyperbolic space through the hyperbolic multimodal semantic alignment (HMSA) module, highlighting the hidden non-Euclidean information in the features. We use hyperbolic distance as the similarity metric for image–text contrastive learning to better integrate the semantic relationships between image–text pairs. The hierarchical nature of hyperbolic space also enhances the recognition ability for similar semantics. In the decoding stage, we design a multimodal decoder to strengthen the semantic relationships between image–text feature pairs and propose two structures to accommodate both BCD and SCD tasks.

In summary, the main contributions of this article are as follows.

1) Designed an HTP generation method for generating semantic text cues that can be applied to different change detection datasets. This method constructs hierarchical text cue information tailored to the SCoI in various datasets.
2) Designed a HMSA module, which achieves semantic alignment between image and text features in hyperbolic space and deeply explores the non-Euclidean characteristics of images.
3) Designed a multimodal decoder architecture that can be adapted for both BCD and SCD tasks by modifying the output branches.
4) Proposed a change detection network based on hyperbolic vision–language representation. Compared to other advanced methods, our approach achieves superior results on both tow BCD datasets (CDD and LEVIR-CD) and an SCD dataset (SECOND).

The rest of this article is organized as follows. The related works are reviewed in Section II. A detailed description of the proposed method is provided in Section III. The experimental results are reported in Section IV. Discussions of key issues are given in Section V. Conclusions are included in Section VI.

II. Related Work

A. Multimodal Representation Learning for RSCD

RSCD has long been regarded as a special semantic segmentation task. However, RSCD fundamentally differs from semantic segmentation: the target semantics of RSCD—namely "changes"—are inherently uncertain. Consequently, studies delving into semantic information specific to RSCD tasks remain relatively uncommon. With the rise of DL methods, RSCD tasks have primarily followed the "pretraining + fine-tuning" paradigm [31]. This approach leverages the rich prior knowledge of pretrained models to mitigate the influence of diverse semantics to some extent. Based on this paradigm, pixel- and object-based methods have been developed. The introduction of FCNs, which allow for end-to-end training on images of arbitrary sizes, shifted the focus toward scene-based methods. The adoption of the

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Siamese network structure by Caye Daudt et al. [15], featuring a shared-parameter dual-branch design tailored to RSCD's dual-temporal input characteristics, has made this architecture the most popular baseline model. However, these methods face two primary limitations. First, they rely heavily on pretrained models trained on the ImageNet dataset [32], which are not fully suited for DL-based remote sensing image analysis due to significant differences in imaging angles, resolutions, and other properties between remote sensing images and natural images in ImageNet [26]. Second, solely capturing image features from training data can limit the model's semantic understanding of remote sensing images. Although recent research [33], [34], [35] have trained pretrained models specifically for remote sensing data using large-scale datasets, issues such as insufficient training data still result in performance gaps compared to classic pretrained models.

In recent years, many researchers have focused on self-supervised learning and multimodal representation learning. As a representative work combining self-supervision and multimodality, CLIP has found extensive applications. The most notable feature of the CLIP model is its ability to generate embeddings for different modalities through specific modality encoders. Compared to purely image-based features, multimodal data provide richer and more accurate semantic information [36], significantly enhancing the model's semantic understanding. CLIP is pretrained on a dataset comprising billions of image–text pairs, including not only natural images but also remote sensing and medical images. It demonstrates impressive zero-shot task performance and is better suited for remote sensing data processing than previous pretrained models. The advantages of multimodal learning are also evident in remote sensing image interpretation. For example, Zhang et al. [37] combined hyperspectral and radar data for hyperspectral classification, while Wang et al. [38] used radar point clouds to supplement the 3-D information absent in remote sensing images, improving semantic segmentation performance. In the context of RSCD tasks, Dong et al. [26] were the first to introduce the multimodal CLIP model, leveraging its strong zero-shot classification capability to extract semantic textual information from CD datasets, thereby enhancing image semantics. Zhao et al. [39] integrated CLIP with the SAM [40] model to segment changes with specified semantics, exploring the distinctions and commonalities of different "change" semantics in CD tasks. Tan et al. [41] used CLIP to extract image semantics in segments, reducing false change phenomena under unsupervised conditions. This article follows the trend of multimodal representation learning by designing a semantic prompting framework to capture the semantic information of remote sensing images and further explores the relationships among multimodal features in hyperbolic space.

B. Application of Hyperbolic Space in DL

Hyperbolic space is a Riemannian manifold with constant negative curvature. Unlike Euclidean space, which has zero curvature, hyperbolic space has the unique ability to represent hierarchical structures. For data with hierarchical characteristics, their structure can be naturally preserved and reconstructed in hyperbolic space. Most existing visual DL methods are defined in Euclidean space. This is partly because, while the real world exists in a curved space, most image data are flattened into a Euclidean form. Additionally, Euclidean space is isomorphic to a vector space, which facilitates the computation of vectors and matrices required by neural networks [42]. In contrast, building DL operations in hyperbolic space requires adhering to manifold constraints. Computational operations on vectors must be redefined within a gyrovector space, a mathematical structure that reflects the properties of the hyperbolic manifold.

Hyperbolic space has been widely used in the early stages of DL development due to its powerful ability to model hierarchical data structures, such as text data, trees, and graphs. These applications have been prominent in fields like natural language processing (NLP), graph neural networks, and knowledge graphs. However, it is only in recent years that researchers have begun to explore the potential of modeling computer vision tasks in non-Euclidean spaces [43]. For remote sensing interpretation tasks, remote sensing images, as a representation of the real world, are often flattened in Euclidean space. This flattening hides certain non-Euclidean features, making them difficult for Euclidean networks to learn. Moreover, studies have shown that in some applications, expert-constructed dissimilarity metrics often exhibit non-Euclidean behavior [44]. Additionally, even in low-dimensional embedding spaces, hyperbolic models can produce high-quality representations, making them particularly advantageous in low-memory and low-storage scenarios [45]. Zhang et al. [46] have shown that the hyperbolic distance in hyperbolic space can be used as a measure of uncertainty. By leveraging this capability to quantify uncertainty or smooth decision boundaries, the robustness of neural networks can be significantly enhanced. There are five commonly used hyperbolic space models [47]. Among them, hyperbolic neural networks and hyperbolic graph neural networks are suitable for textual data, while hyperbolic mixture models are still not mature. For computer vision tasks, two models are commonly used: the Poincaré ball model and the Lorentz model. These two models are implemented similarly; however, the Lorentz model is more stable than the Poincaré ball model and is better suited for multimodal inputs. Therefore, in this work, we adopt the Lorentz model to implement hyperbolic space mappings.

In the context of RSCD tasks, Yang et al. [48] integrated bitemporal features in hyperbolic space and demonstrated through experiments on multiple datasets the necessity of hyperbolic information for RSCD tasks. In multimodal representation learning, Desai et al. [28] extended the CLIP model to enable contrastive learning of multimodal features in hyperbolic space, providing evidence for the effectiveness of hyperbolic space in modeling image–text relationships.

III. Methods

A. Overview

This section provides a detailed description of our change detection network, HyperCLIP-CD, which is based on a Siamese structure integrating visual–language representation learning methods. The network framework, as shown in Fig. 2,

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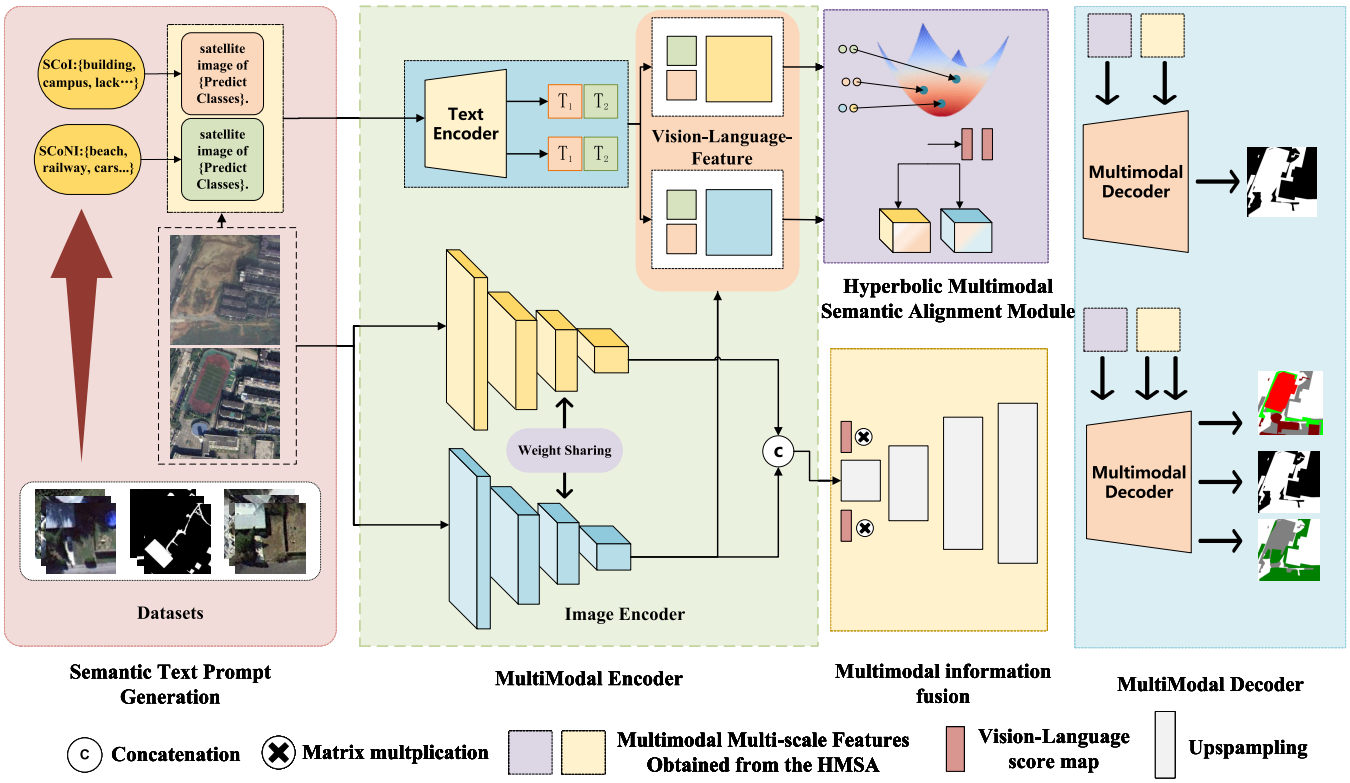


Fig. 2. Overall architecture of the proposed HyperCLIP-CD. Multimodal multiscale features refer to the outputs of the HMSA module and the output of the multimodal fusion operation in the multimodal decoder. The vision–language score map is an embedding containing multimodal semantic information generated during the operations of the HMSA, producing a set for each temporal phase.

leverages the powerful zero-shot classification capability of the CLIP pretrained model. We construct textual modality data via a tailored prompt generation strategy, in which prompts are categorized into foreground semantics (SCoI) and background semantics (SCoNI). Specifically, for each image, the top three semantic categories predicted by the pretrained CLIP model are selected. Based on the aggregated semantic predictions across the dataset, the most frequently occurring categories are defined as SCoI, while the remaining categories are grouped as SCoNI. For the fusion of image and text information, the network extracts multimodal features by combining Siamese structure-based image and text encoders. To delve into non-Euclidean information and align multimodal semantics, we introduce the HMSA module. This module projects multimodal features into hyperbolic space, utilizes hyperbolic distance to measure semantic similarity, and enhances multimodal semantic coherence. In the final decoder stage, tailored to the distinct requirements of BCD and SCD, we capitalize on the visual–linguistic features obtained in the encoding phase to design two decoding modes to address diverse task demands.

TABLE I
HTP

Root node	primary node	secondary node
Man-made Structures	Buildings	Church, Cabin, Dense residential, Single-family residential, Building
	Commercial & Industrial	Stadium, campus, Park, square, parking lot, storage tanks, container, oil tank, Commercial area, Industrial area
	Transportation	Airport, Runway, Freeway, Highway, Road, Intersection, Bridge, Railway, Interchange, Cars, Ship, Airplane
	Sports	Basketball court, Golf course, Stadium
Natural Environments	Environments	Mountain, Desert, Island, Mining, Forest, Fertile land, Tree
	Water Bodies	Lake, River, Sea, Pond, Wetland
	Land	Bare land, Impermeable surface, Terrace, Shrubbery, Snow land, Meadow, Prairie, Farmland, Cotton field

B. HTP Generation

Before delving into multimodal representation learning, we first generate textual prompts based on the CLIP pretrained model. Research by Desai et al. [28] suggests the presence of an intuitive hierarchical structure between vision and language, where a textual description encompasses a broader semantic scope compared to an image. As textual descriptions become more precise, the similarity between vision and language increases. The hyperbolic space utilized in this study possesses unique geometric properties, with distances from a node to the root node exponentially increasing. This property aligns well with embedding multimodal semantics, where accurate textual semantics closely approximate image

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semantics, while irrelevant semantics diverge. Hence, we design hierarchical textual descriptive prompts for subsequent methods, as outlined in Table I. Specifically, we hierarchically classify common remote sensing land-cover categories for more precise descriptions. Even if the CLIP pretrained model misclassifies, as long as the root node of the classification aligns with the true semantics, significant semantic shifts affecting subsequent computations are avoided.

Moreover, while generating textual prompts solely from image data can enhance the model's semantic comprehension, a plethora of irrelevant semantics can disrupt the model's recognition of changing regions to some extent, leading to the misidentification of pseudo-change regions. To delineate the current dataset's SCoI, we employ the dataset's actual labels as masks to segment the corresponding change regions between time phase A and time phase B. The computation steps are as follows:

$$F_{(A,B)}^{\text{mask}} = \eta(F_G) \cdot F_{(A,B)} \tag{1}$$

where $F_{(A,B)}$ represents the original images of time phases A and B, F_G denotes the true labels corresponding to A and B, η is the normalization function, and $F_{(A,B)}^{\text{mask}}$ denotes the region of interest in time phases A and B.

We generate textual prompts based on $F_{(A,B)}^{\text{mask}}$. Specifically, for each image in the dataset, the top three semantic categories with the highest CLIP prediction scores are selected. These semantic categories from all images are then aggregated, and their occurrence frequency is computed as the number of times a category appears among the top-three predictions across the entire dataset. The semantic categories are ranked according to their dataset-level occurrence frequency, and the top two-thirds of the ranked categories are defined as the SCoI for the dataset, while all remaining categories are unified as the SCoNI. The reason for not categorizing all classes in $F_{(A,B)}^{\text{mask}}$ as interesting is that not all semantic categories in the labels are truly worth attention. For instance, if the highest predicted categories for one pair in $F_{(A,B)}^{\text{mask}}$ are "building" → "tree" and for another pair are "building" → "grass," the semantic focus of this change detection task might only be "building," disregarding the specific changed semantics. In such cases, the proportion of the truly focused semantics in $F_{(A,B)}^{\text{mask}}$ is likely to be significantly higher. It is worth mentioning that since the SCD task typically specifies the semantic classes of interest, we regard all the specified classes as SCoI, while the semantics in our hierarchical textual descriptions that are unrelated to these SCoI are treated as noninterest semantics. Since the SCD task typically specifies the semantic classes of interest, we regard all the specified classes as SCoI, while the semantics in our hierarchical textual descriptions that are unrelated to these SCoI are treated as noninterest semantics.

Ultimately, we select the top 5 classes with the highest confidence from both SCoI and SCoNI (filling any shortfall with empty classes), generating two sets of textual descriptions using the format "satellite image of Predict Classes." We feed the SCoI prompt P_1 and the SCoNI prompt P_2 into the text encoder.

C. Hyperbolic Multimodal Semantic Alignment Module

The encoding end of HyperCLIP-CD consists of parallel and parameter-sharing image encoders and two sets of text encoders, following the CLIP architecture, where different backbone networks can be flexibly adopted. In this work, we implement and evaluate two commonly used image encoders, namely ResNet-50 and ViT-B/16, to investigate their effectiveness within the proposed framework. Both encoders are integrated in a unified manner without modifying the overall network structure or training strategy. The text encoder in this study adopts a Transformer architecture.

The Siamese image encoder can extract two sets of bitemporal features, F_A^1– F_A^4 and F_B^1– F_B^4, at different scales. Typically, shallow features contain more coarse-grained features and edge information, while deeper features encompass finer-grained features and semantic information. We utilize the deepest features, denoted as F, as the image semantic embedding. This embedding aligns with the text embedding in our hyperbolic multimodal semantic alignment module, as illustrated in Fig. 3. Before this alignment, attention pooling on F is required to capture key semantic features of the image, reshape them into image embeddings, and the specific process is outlined as follows:

$$V_e = reshape\left(softmax\left(\frac{(Q_i \cdot K_i)}{\sqrt{(d_k)}}\right) V_i\right) \tag{2}$$

where $V_e \in \mathbb{R}^{C\times HW}$ denotes the image feature embedding sequence obtained by attention pooling, with C representing the channel dimension and $H \times W$ indicating the spatial resolution of the feature map. In the equation, Q, K, and V denote query, key, and value matrices, respectively. The subscript i indexes the spatial tokens obtained by flattening the feature map $F \in \mathbb{R}^{C\times H\times W}$ into a sequence of length HW, where each token corresponds to a spatial position in the feature map. Specifically, we set $Q_i = V_i = F$ and $K_i = F^T$, where F represents the deep feature extracted by the image encoder, containing rich fine-grained and high-level semantic information. The scaling factor d_k is introduced to normalize the dot-product attention and stabilize the training process.

Each branch of the dual-path text encoder receives inputs of two textual prompts, namely SCoI prompt P_1 and SCoNI prompt P_2. After text feature extraction through the Transformer, two text feature embedding sequences, T_1 and T_2, are output. $T \in \mathbb{R}^C$. Subsequently, T_1 and T_2 are combined with the preceding image embedding sequences in the hyperbolic multimodal semantic alignment module. To achieve this, we first project the image–text sequences into hyperbolic space. Specifically, we normalize the embedding vectors to reduce differences and then apply a transformation to map the resulting vectors into hyperbolic space. In hyperbolic geometry, we often use exponential operations to realize this space mapping. The specific implementation is as follows:

$$y = \exp_\lambda^r(\eta(v)) \tag{3}$$

where $y, o \in D_c^d$, with D_c^d representing a hyperbolic surface, λ as the anchor point, and $1/r$ as the hyperbolic radius. Moreover, $v \in R^c \cong T_x D_c^d$ denotes a vector in Euclidean space. Here, v can simultaneously refer to the image feature sequence

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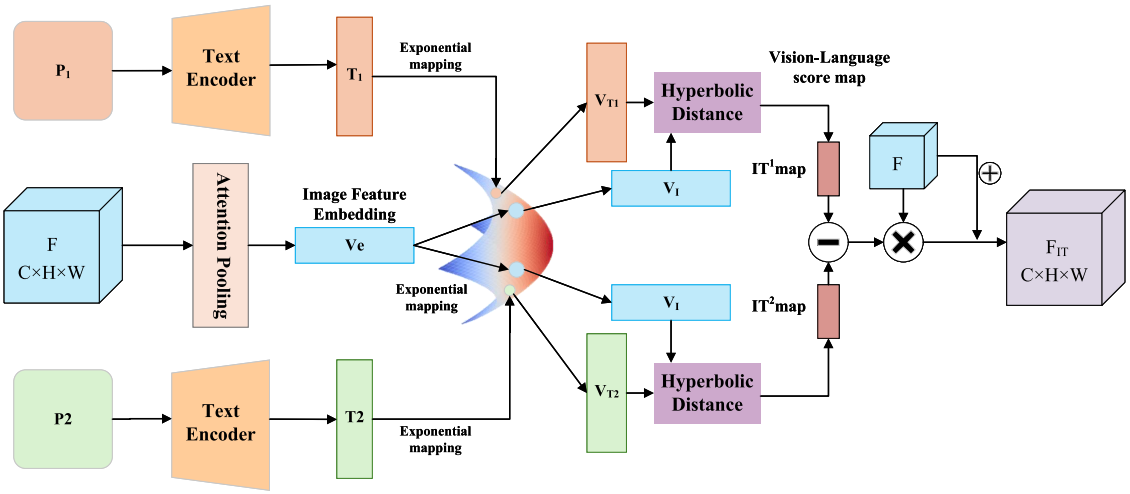


Fig. 3. Illustration of the hyperbolic multimodal semantic alignment module, where F is the deepest feature of the image encoder, T1 and T2 are the corresponding SCoI and SCoNI text features, and the exponential mapping operation is used to map features onto the hyperbolic space.

V_e and the text feature sequence T_i. ${}_xD_c^d$ symbolizes the tangent space of the hyperbolic surface at point x, acting as an intermediary station for the conversion between Euclidean and hyperbolic spaces. η represents the normalization operation.

The specific form of the exponential mapping depends on the hyperbolic model used. In this article, we adopt the Lorentz model $\mathcal{L}^c \in R^{c+1}$, for which the exponential mapping is defined as follows:

$$exp_\lambda^r(v) = \cosh\left(\sqrt{r}\|v\|\right)\lambda + \frac{\sinh\left(\sqrt{c}\|v\|\right)}{\sqrt{c}\|v\|}v \tag{4}$$

where $\|\cdot\|$ represents the L2-norm.

We extend the dimension of vector v to align with the hyperbolic space dimensionality, resulting in $v = [v, 0] \in \mathbb{R}^{c+1}$. At this point, v belongs to the tangent space at the origin of the hyperbolic surface, with the anchor point λ being the origin. Consequently, we can simplify the exponential mapping in (4) by solely parameterizing the spatial components of the Lorentz surface

$$exp_0^r(v) = \frac{\sinh\left(\sqrt{c}\|v\|\right)}{\sqrt{c}\|v\| + eps}v \tag{5}$$

where eps is a small value added to prevent division by zero.

The specific operational implementation to obtain hyperbolic embeddings for the image and two sets of text is as follows:

$$V_{I/T} = MT\left(exp_0^r\left(v_{I/T}\right)\right) \tag{6}$$

where $V_{I/T}$ denotes the hyperbolic gyrovector embeddings for the image and text and $MT()$ represents the construction of a manifold tensor. The hyperbolic gyrovectors must be sequentially stored using the manifold tensor.

We utilize the hyperbolic distance between gyrovectors embedded on the Lorentz surface as a metric to construct the vision–language score map IT_{map}. This score map explicitly represents the semantic relationship between image and text feature sequences and can be regarded as a multimodal embedding with textual semantics for subsequent computations. The specific implementation is as follows:

$$d_{\mathcal{L}}(x, y) = \sqrt{\frac{1}{r}} \cdot \cosh^{-1}\left(1 + \frac{2\|x - y\|^2}{\left(1 - \|x\|^2\right)\left(1 - \|y\|^2\right)}\right) \tag{7}$$

$$M_{IT}^i = \frac{d_{\mathcal{L}}(x, y)}{\sqrt{c}} \tag{8}$$

$$IT_{map}^i = logmap\left(Softmax\left(expmap\left(M_{VT}^i\right)\right)\right) \tag{9}$$

where $d_{\mathcal{L}}(x, y)$ denotes the geodesic distance on the Lorentz surface and M_{IT}^i is the image–text feature similarity matrix for i = (1, 2), representing the similarity between the image features and SCoI and SCoNI, respectively. $expmap()$ signifies the exponential mapping, while $logmap()$ denotes the logarithmic mapping, used to remap the image–text score map back to Euclidean space for subsequent computations. At this point, IT_{map}^1 and IT_{map}^2 obtained represent the semantic information of interest and disinterest matched with the image features. To further highlight the role of semantic interest and suppress pseudo-changes, we proceed with a subtraction operation on the two score maps

$$IT_{map} = Norm\left(IT_{map}^1 - IT_{map}^2\right) \tag{10}$$

norm in the above equation represents a normalization function. Ultimately, $IT_{map} \in R^{C\times HW}$, and the original image features F are multiplied element-wise with IT_{map} to integrate textual semantic information and hyperbolic information, yielding new features F_{IT}. As a precautionary measure, we also add the original image features F as a residual branch to the new features. The specific implementation is as follows:

$$F_{IT} = \left(F \times IT_{map}\right) \oplus F \tag{11}$$

where $\oplus$ denotes element-wise summation.

We feed the image features full of textual semantics and hyperbolic information and the score map IT_{map}^1 containing SCoI features into the multimodal decoder.

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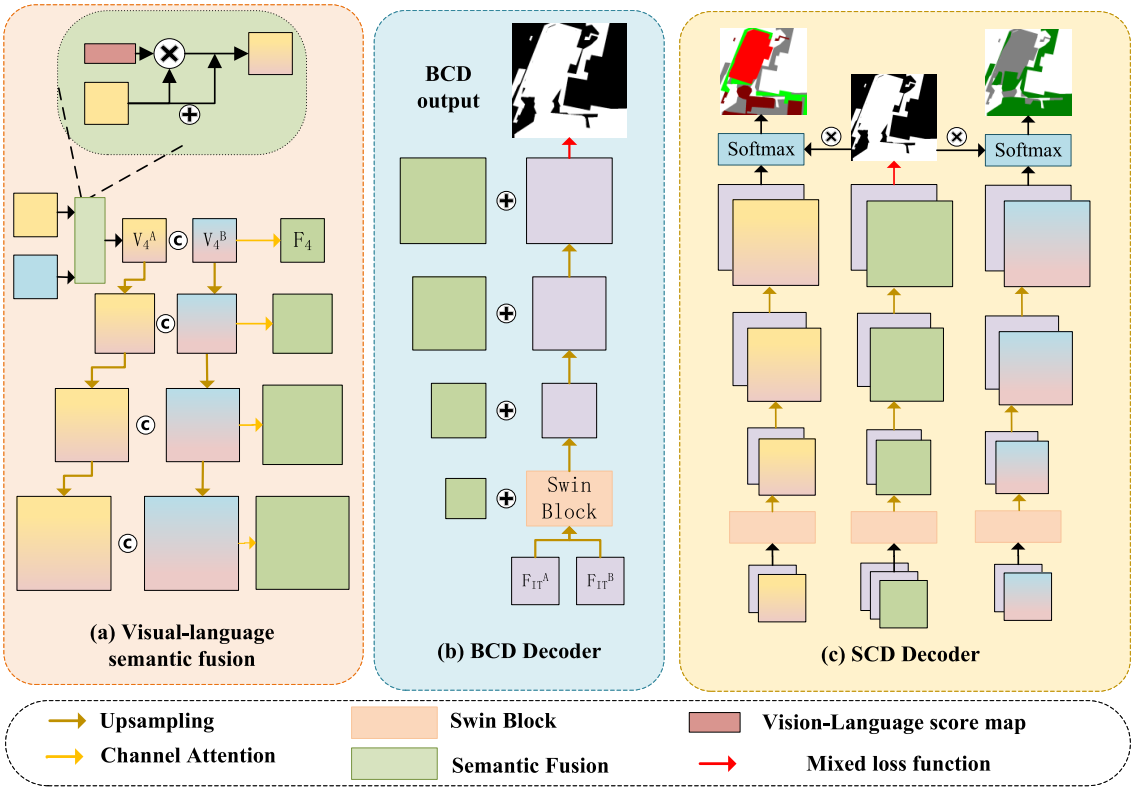


Fig. 4. Illustration of the multimodal decoder. (a) Combines the features in the image encoder with the multimodal embedding and generates a multiscale feature map. (b) BCD decoder, which uses the multiscale feature map and the output of HMSA to obtain the change map. (c) SCD decoder, which combines multiple sets of features and outputs the change and semantic segmentation results of the two phases.

D. Multimodal Decoder

To better utilize the multimodal features extracted during the encoding phase, we have designed a multimodal decoder. There are two different architectures for the BCD and SCD tasks, as depicted in Fig. 4. Let us first introduce the BCD architecture.

To further enhance the decoder's global attention capability, we employ Swin Transformer blocks as the basic decoding units. Based on the multiscale image features F^1–F^4 extracted in the encoding stage and the textual semantic information of interest IT_{map}, we perform multiscale visual–semantic feature fusion on the two branches, supplementing the semantic features in the decoding phase. We then merge the dual-branch features at each scale to obtain a set of multiscale features that integrate multimodal information across multiple time phases. The specific implementation is as follows:

$$V_j^4 = F_j^4 + \left(F_j^4 \times IT_{map}\right) \tag{12}$$

$$V_j^i = Upsample\left(F_j^{i-1}\right) \tag{13}$$

$$F_i = ChannelAttention\left(concat\left(V_A^i, V_B^i\right)\right) \tag{14}$$

where V_j^i represents the fused semantic features of time phase j at layer i, F_i denotes the fused bitemporal feature at layer i, and the channel attention operation highlights the most relevant information in the concatenated bitemporal features. Subsequently, we concatenate and decode the two sets of hyperbolic multimodal features F_{IT} obtained in Section III-A, merge them with the corresponding F_i at each layer, upsample them until restoring the original image size, and obtain the output

$$D_4 = f\left(F_{IT}^A, F_{IT}^B\right) \oplus F_4 \tag{15}$$

$$D_i = Upsample\left(D_{i+1}\right) \oplus F_i \tag{16}$$

where D_i represents the different scale features obtained in the decoder stage, f denotes the Swin Transformer decoding block, and the final prediction structure output corresponds to the upsampled result of D_1.

Unlike the BCD task that focuses solely on "where the change is," the SCD task aims to determine "what has changed." This requires not only outputting a binary image of the changed areas but also producing semantic segmentation maps of the changed regions in two time phases. Therefore, the decoder structure for the SCD task differs slightly from the BCD task, as shown in Fig. 4(b). The operations are similar to the BCD decoding, with the major distinction being the three-branch decoder output. The intermediate binary change area branch aligns with the BCD setup, while the semantic segmentation branches for the two time phases combine the image semantic fusion features V_j^i with the hyperbolic multimodal features F_{IT}^j and upsample them. The specific operations are as follows:

$$SD_j^4 = f\left(concat\left(V_j^4, F_{IT}^j\right)\right) \tag{17}$$

$$SD_j^i = V_j^i \oplus Upsample\left(SD_j^4\right) \tag{18}$$

$$output = BCD_{mask} \otimes Softmax\left(SD_j^1\right) \tag{19}$$

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where SD_j^i represents the output of the semantic decoder at layer i, where the deepest layer combines hyperbolic visual–linguistic information, then upsamples while merging with lower-level features with higher resolution. $Softmax$ is applied at the shallowest layer for semantic classification of image pixels, with the final binary change result BCD_{mask} serving as the mask to segment the semantic map of the changed areas.

E. Loss Function

For the BCD task, the final output is a binary image. Due to the fact that the changed regions are typically much fewer in comparison to the unchanged regions, change detection data often exhibits class imbalance. Therefore, we utilize a hybrid loss function, combining the commonly used Dice loss for segmentation tasks with the weighted cross-entropy loss tailored for addressing class imbalance issues. Our loss function is defined as follows:

$$L = L_{dice} + L_{wce}. \tag{20}$$

The Dice loss and the weighted cross-entropy loss are, respectively, defined as follows:

$$L_{dice} = 1 - \frac{2 \cdot Y \cdot softmax\left(\hat{Y}\right)}{Y + softmax\left(\hat{Y}\right)} \tag{21}$$

where Y represents the ground truth and $\hat{Y}$ is the change map

$$L_{wce} = \frac{1}{H \times W} \sum_{k=1}^{H \times W} \text{weight}\left[\text{class}\right] \cdot \left(\log\left(\frac{\exp\left(\hat{y}\left[k\right]\left[\text{class}\right]\right)}{\sum_{l=0}^{1} \exp\left(\hat{y}\left[k\right]\left[l\right]\right)}\right)\right). \tag{22}$$

For the SCD task, in addition to optimizing the BCD results, we need to optimize the semantic segmentation tasks for land cover in two bitemporal phases. We also utilize the weighted cross-entropy loss from (21) for optimization. The final definition of the SCD loss function is

$$L_{SCD} = L_{wce}^{BCD} + \frac{1}{2}\left(L_{wce}^{T1} + L_{wce}^{T2}\right). \tag{23}$$

IV. Experiments

We conducted numerous comparative experiments to evaluate the performance of HyperCLIP-CD. Through ablation studies, we validated the effectiveness of our proposed text prompt generation method and hyperbolic multimodal semantic alignment module.

A. Experimental Setup

1) Datasets: To demonstrate the generalizability of our method to the RSCD task, we selected three significantly different and representative datasets, including the multitype change dataset CDD [49] and the building change dataset LEVIR-CD [50], both of which are datasets for the BCD task. Additionally, we validated our method on the classic SCD dataset SECOND [51].

a) CDD: The dataset comprises satellite images captured by Digital Globe at different seasons, with spatial resolutions ranging from 3 to 100 cm/px. The original dataset was processed through cropping and rotation to generate 16 000 pairs of images sized 256 × 256 pixels each. The final dataset was divided into three parts: 10 000/3000/3000 for training/validation/test, respectively. This dataset includes various types of changes, making it a classic dataset for evaluating the performance of change detection models in different environments.

b) LEVIR-CD: The dataset contains 637 pairs of high-resolution remote sensing images from Texas, USA, each image sized 1024 × 1024 with a resolution of 0.5 m/px, captured between 2002 and 2018. Used for building change detection, it covers various types of buildings. To align with recent CD literature, we segmented each image pair into patches of size 256 × 256 pixels, resulting in 10 192 images for training and testing, with splits of 7120/1024/2048 for training/validation/test, respectively.

c) Second: The dataset consists of aerial images from cities like Hangzhou, Chengdu, and Shanghai, totaling 4662 pairs, each image sized 512 × 512 with a resolution of 0.53 m/pixel. It classifies six main land-cover categories and features detailed annotations, including nonvegetated ground surfaces, trees, low vegetation, water, buildings, and playgrounds. This dataset uses land-cover maps and nonchange masks to represent change areas, with changed pixels accounting for 19.87% of the total pixels. We used 1/5 of the data (593 pairs of RSIs) as the test set, with the remaining 2375 pairs used for training.

2) Implementation Details: HyperCLIP-CD is implemented based on the PyTorch framework. We conducted experiments on a machine equipped with an NVIDIA RTX A5000 GPU, training the model for 80 epochs to ensure full convergence. To minimize the impact of external factors on the experiments, we only employed data augmentation techniques like RandomFlip and PhotoMetricDistortion. During training, the batch size for input images was set to 8, the Adam optimizer was utilized with a learning rate of $1e^{-4}$, and the learning rate was decayed by 0.7 every 5 epochs. The weights of each convolutional layer were initialized using KaiMing normalization.

Before training the model, we utilized the change semantic text prompt generation method proposed in this article to generate both interested and disinterested text based on the dataset being used. We selected the top three semantically confident texts for each prompt and fed them into the network. From our tests, this operation did not significantly increase computation time or impact training efficiency.

3) Evaluation Metrics: For the BCD task, we evaluate the model performance using five commonly used metrics in change detection tasks: precision (Pre), recall (Rec), $F1$-score (F1), intersection over union (IOU), and overall accuracy (OA). The calculation methods for these metrics are as follows:

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{24}$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{25}$$

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TABLE II
PERFORMANCE COMPARISON ON THE CDD DATASET

Type	Methods	OA	Pre	Rec	F1	IOU
CNN-base	DASNet	97.89	91.41	92.45	91.92	87.97
	FDCNN	97.16	87.90	86.99	87.44	86.42
	SNUNet	98.78	95.61	94.05	94.83	90.12
	STANet	97.52	88.98	93.11	91.00	87.61
	USSFC-Net	98.69	93.65	96.08	94.74	90.06
	HES-NET	99.42	97.67	96.77	97.22	95.57
Transformer-base	BIT	98.97	94.01	96.63	95.26	91.08
	HMCNet	99.09	**98.66**	92.71	95.55	93.75
	ChangFormer	99.14	96.24	96.23	96.37	94.10
	TransUNetCD	99.22	96.63	96.69	96.79	95.17
	ScratchFormer	**99.46**	-	-	97.77	95.81
Mamba-base	RSM-CD	99.29	96.30	97.33	96.81	93.82
	ChangeMamba-Base	99.33	96.60	96.79	96.64	93.61
	CDMamba	99.35	97.65	97.08	97.36	94.86
CLIP-base	ChangeCLIP(VIT/16)	99.24	96.78	97.28	97.02	94.18
	ChangeCLIP(ResNet50)	99.31	96.96	97.16	97.06	94.29
	Ours(VIT/16)	99.32	98.16	97.29	97.72	94.54
	Ours(ResNet50)	99.37	98.05	**97.70**	**97.87**	**95.84**

$$\mathrm{F1} = \frac{2 \times \mathrm{Pre} \times \mathrm{Rec}}{\mathrm{Pre} + \mathrm{Rec}} \tag{26}$$

$$\mathrm{IOU} = \frac{TP}{TP + FN + FP} \tag{27}$$

$$\mathrm{OA} = \frac{TP + TN}{TP + TN + FN + FP} \tag{28}$$

where false negative (FN) represents the number of pixels incorrectly classified as unchanged, false positive (FP) represents the number of pixels incorrectly classified as changed, and true positive (TP) is the number of pixels correctly classified as changed. The $F1$-score and IOU provide a comprehensive reflection of the precision and recall levels and serve as the primary criteria for assessing model quality.

For the SCD task, we employ four common metrics: OA, mean IOU (mIOU), separation Kappa coefficient (SeK), and $F1$-score. While the metrics differ slightly in their implementation for SCD compared to BCD, they share the same essence. Refer to [52] and [53] for specific definitions. SeK is a unique metric for the SCD task, assessing the effectiveness of semantic classification and serving as a primary criterion for evaluating SCD model performance.

B. Comparison Experiments

1) Comparison Methods: We compared our experimental results with 21 representative DL-based CD methods, comprising 16 methods for the BCD task categorized based on their model architectures into CNN-based methods: STANet [50], FDCNN [54], DASNet [55], SNUNet [14], USSFC-Net [56], and HES-Net [48]; Transformer-based methods: BIT [18], ChangeFormerV6 [17], HMCNet [57], TransUNetCD [58], and ScratchFormer [59]; Mamba-based methods: RSM-CD [21], ChangeMamba-Base [1], and CDMamba [20]; and CLIP-based methods: ChangeCLIP (comparing versions based on VIT/16 and ResNet50) [26] and LGCD [60].

For the SCD task, we compared against six novel methods: HRSCD (selecting the best-performing str.4 version in this study) [61], SCDNet [62], SSCD-1 [52], TED [53], ScanNet [53], and SMNet [63].

To ensure fairness, we replicated the results of open-source methods using the same data settings and experimental environments. For methods without open-source code, we relied on the results reported in their original papers.

2) Comparison on the BCD Datasets: As shown in Tables II and III, we extensively evaluated HyperCLIP-CD on two BCD datasets (CDD and LEVIR-CD). Quantitative analysis reveals that our method achieves state-of-the-art performance in all data sets. The symbol '–' denotes missing results from the original publications. In particular, HyperCLIP-CD achieves notable improvements in IoU, which is especially sensitive to boundary quality and semantic consistency.

On the CDD dataset, HyperCLIP-CD equipped with a ResNet-50 backbone attains an IoU of 95.84%, outperforming all comparative methods. It exceeds ChangeCLIP, which also adopts a ResNet-50 encoder, by 1.55%. Meanwhile, the $F1$-score reaches 97.78%, surpassing ChangeCLIP by 0.81%, which is a meaningful gain given the already high baseline performance on this dataset. When using a ViT-B/16 backbone, HyperCLIP-CD also achieves competitive results, with only marginal performance differences compared to ResNet-50, demonstrating the robustness of the proposed framework to backbone selection. Compared with the recently popular Mamba-based methods, HyperCLIP-CD also achieves comprehensive improvements across all metrics. These metrics show the precision of HyperCLIP-CD in complex object detection data. Visual analysis in Fig. 5 demonstrates minimal FPs and negatives compared to other methods, with precise identification even in ambiguous detail regions. The hierarchical structure in hyperbolic space significantly aids in classifying complex scenes, exhibiting robust noise resistance and detail detection capabilities.

On the LEVIR-CD dataset, HyperCLIP-CD demonstrates similarly strong performance in building change

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TABLE III
PERFORMANCE COMPARISON ON THE LEVIR-CD DATASET

Type	Methods	OA	Pre	Rec	F1	IOU
CNN-base	DASNet	96.52	84.15	89.78	86.87	78.18
	FDCNN	97.58	86.14	89.39	87.73	79.03
	SNUNet	98.41	89.43	87.72	88.57	79.72
	STANet	98.92	92.49	85.80	89.02	80.20
	USSFC-Net	98.95	88.79	91.17	89.96	82.42
	HES-NET	99.01	90.74	90.33	90.53	83.05
Transformer-base	BIT	98.95	90.65	87.53	89.48	80.86
	HMCNet	99.07	91.68	89.82	90.74	83.05
	ChangFormer	99.04	90.83	90.18	90.50	82.66
	TransUNetCD	99.11	92.43	89.82	91.11	83.67
	ScratchFormer	99.16	-	-	91.11	83.67
Mamba-base	RSM-CD	99.04	92.52	89.73	91.10	83.66
	ChangeMamba-Base	99.01	91.59	88.78	90.16	82.09
	CDMamba	99.06	91.43	90.08	90.75	83.07
CLIP-base	ChangeCLIP(VIT/16)	99.14	93.68	89.04	91.30	83.99
	ChangeCLIP(ResNet50)	99.16	92.60	90.67	91.62	84.54
	LGCD	99.17	89.47	**92.14**	91.89	-
	Ours(VIT/16)	99.17	**93.85**	89.65	91.70	84.67
	Ours(ResNet50)	**99.20**	93.49	90.78	**92.11**	**85.32**

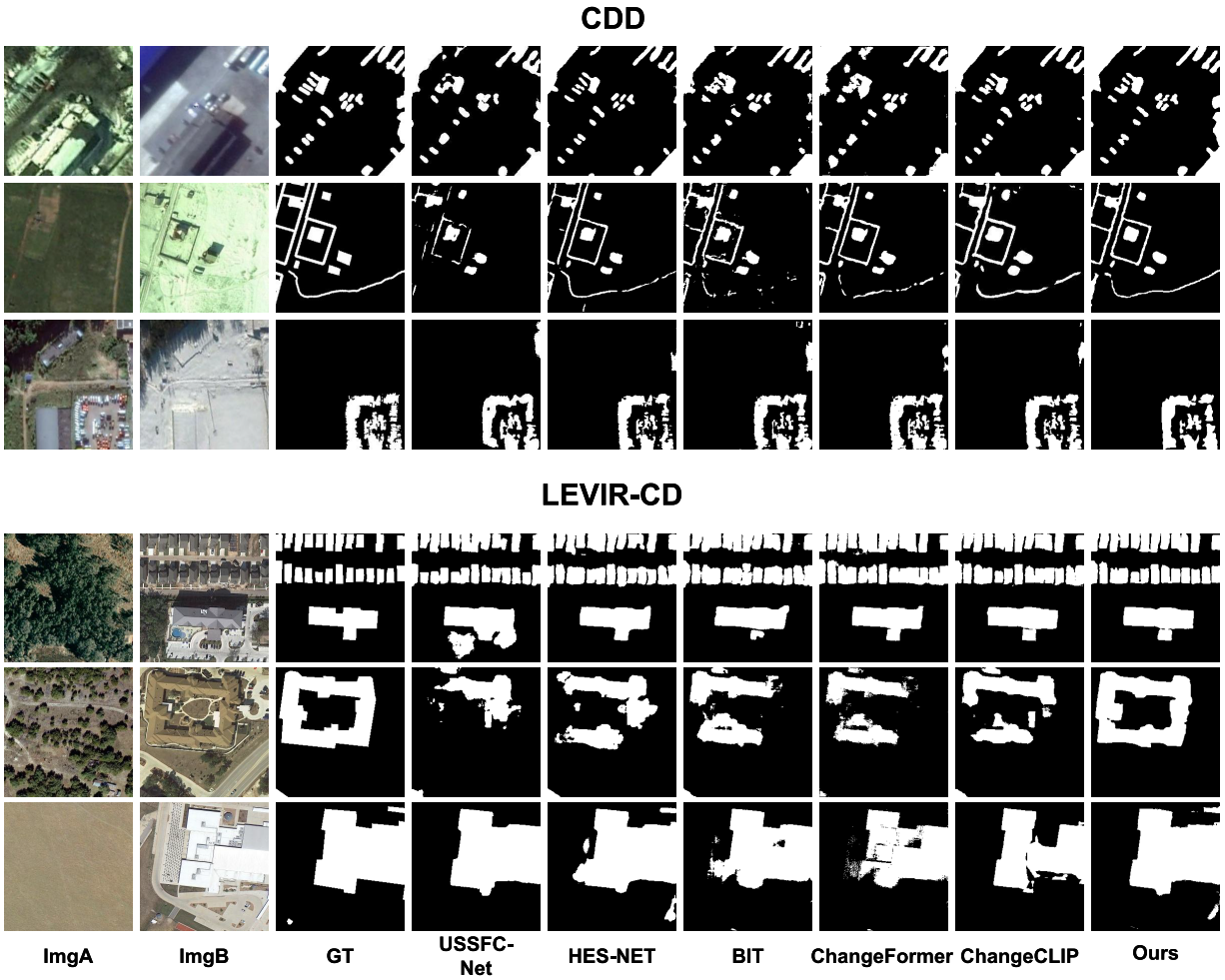


Fig. 5. Visualization comparison of results on two BCD test sets. Here, we take two CNN-based and two Transformer-based methods. ChangeCLIP takes the RESNET50-based version with better performance.

detection. Using the ResNet-50 backbone, the IoU and $F1$-score reach 85.32% and 92.11%, respectively, surpassing CNN-based methods such as HES-Net by 2.27% and 1.58%, and Transformer-based methods such as TransUNetCD by

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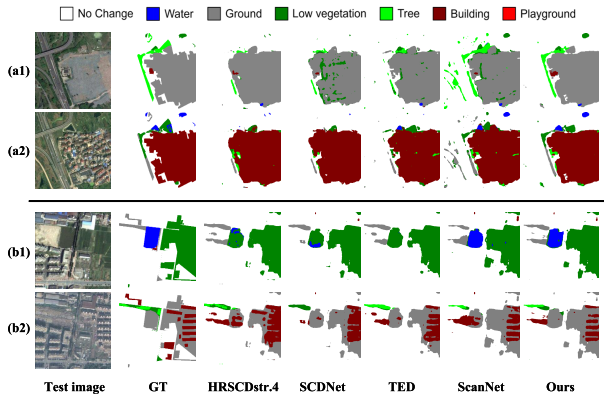


Fig. 6. Visualization comparison of results on the SECOND test sets. White is TP and black is a true negative. (a1) and (b1) Pretemporal images. (a2) and (b2) Posttemporal images.

TABLE IV
PERFORMANCE COMPARISON ON THE SECOND DATASET

Methods	OA	mIOU	F1	Sek
HRSCDstr.4	83.88	70.58	58.21	18.80
SCDNet	84.26	70.97	60.01	19.73
SSCD-1	85.62	72.14	61.22	21.86
TED	85.54	72.19	61.56	22.07
SMNet	84.29	71.62	61.05	20.83
ScanNet	86.13	72.48	61.65	22.29
Ours	**87.36**	**72.95**	**63.18**	**22.94**

1.65% and 1.00%. Although ViT-based encoders are commonly adopted in CLIP-related methods and achieve strong performance in general multimodal tasks, we observe that ResNet-50 yields slightly more stable and consistent results within our hyperbolic alignment framework.

This phenomenon may be attributed to the fact that deep convolutional features obtained through multiple stages of downsampling preserve rich local and hierarchical information, which is beneficial for non-Euclidean representation learning, as also suggested by Yang et al. [48]. In contrast, ViT-based encoders emphasize global context modeling via patch-level tokenization, which may introduce additional variability when coupled with hyperbolic embeddings. It is worth noting, however, that the performance gap between different backbones remains relatively small, and HyperCLIP-CD consistently achieves high accuracy and recall across datasets. Overall, our model displays excellent performance on BCD tasks.

3) Comparison on the SCD Dataset: To demonstrate the versatility of our approach, we conducted relevant experiments on the SCD task as well. The quantitative analysis of the experimental results on the SECOND dataset is presented in Table IV, while the visual analysis is depicted in Fig. 6.

From Table IV, it is evident that our method also achieves satisfactory results on the SECOND dataset. Even when compared with excellent methods specifically designed for SCD tasks, HyperCLIP-CD consistently outperforms in all metrics. Among the comparative methods, TED and ScanNet also incorporate semantic information from the dual-temporal branch features in their model construction, leading to relatively superior metrics compared to other methods. However, the limitation of single-modal image features ultimately restricts the models' ability to explore and utilize semantic information fully. HyperCLIP-CD outperforms ScanNet by 0.65% in SeK and 1.53% in $F1$-score. The visual results also demonstrate that not only is our method proficient in detecting change regions, but it also excels in semantic type recognition compared to other comparative methods.

TABLE V
ABLATION STUDY IN THE $F1$ INDEX. BASELINE: CLIP IMAGE–TEXT ENCODER, MD: MULTIMODAL DECODER

Methods	CDD	LEVIR-CD	SECOND
Baseline	95.82	89.97	60.61
Baseline + HTP(w/o SCoNI)	96.07	90.12	61.25
Baseline + HTP	96.24	90.15	61.33
Baseline + HMSA	96.73	90.94	61.82
Baseline + HTP(w/o SCoNI)+HMSA	96.92	90.89	62.34
Baseline + HTP + HMSA	97.25	91.54	62.71
HyperCLIP-CD(with MD)	**97.87**	**92.11**	**63.18**

C. Ablation Study

1) Module Effectiveness Ablation: To validate the effectiveness of several modules in this study, we conducted a series of ablation experiments using different module combinations on three datasets, with the experimental results presented in Table V. In these ablation experiments, ResNet-50 is adopted as the default image backbone for the proposed method.

As shown in Table V, the ablation experiments exhibit similar trends across the three datasets. Our baseline model employs a Siamese network with a CLIP image–text encoder and directly generates semantic text prompts using the 56 labels from [26]. Replacing the semantic text prompts with the HTP method introduced in this study resulted in slight improvements in metrics for both BCD datasets. Notably, on the SCD dataset SECOND, there was a substantial improvement of 0.72%, highlighting the superiority of constructing semantic prompts using the HTP method.

To further analyze the internal mechanism of HTP, we introduce two additional ablation variants that remove the negative semantic guidance, denoted as "HTP(w/o SCoNI)," with and without the HMSA module. As reported in Table V, in both cases, models incorporating both SCoI and SCoNI consistently outperform their SCoI-only counterparts across all datasets. This observation indicates that negative semantic guidance plays an indispensable role in the proposed framework and is not merely amplified by the HMSA module. The underlying reason lies in the relative nature of CLIP-based similarity: relying solely on positive semantic cues is insufficient to suppress interference from semantically irrelevant yet visually dominant background regions. By explicitly modeling SCoNI, HTP introduces a complementary semantic constraint that enlarges the separation between change-relevant semantics and

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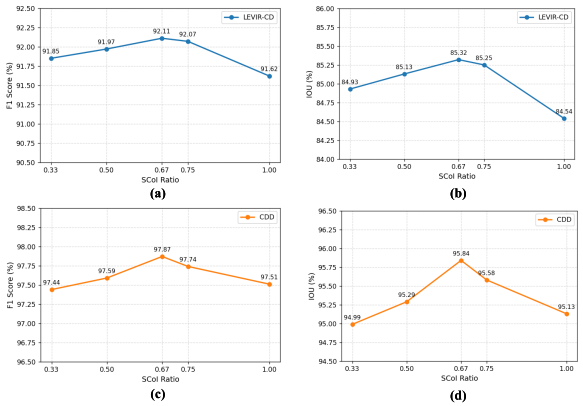


Fig. 7. Visualization of sensitivity analysis for the SCoI acquisition ratio threshold. The x-axis denotes the proportion of semantic categories, ranked by frequency, allocated to SCoI. (a) and (b) Sensitivity results in terms of $F1$ and IoU on LEVIR-CD. (c) and (d) Similar evaluations on the CDD dataset.

stable background semantics in the embedding space, thereby improving robustness and discriminability.

The HMSA module consistently boosted model performance by over 1% on average across the three datasets. This enhancement is attributed not only to the necessity of hyperbolic information for remote sensing image interpretation but also to the synergy between multimodal learning and hyperbolic space operations, which can reduce the disparity between visual-textual features by calculating distances based on semantic similarity.

When combining the HTP and HMSA methods, the model achieved exceptional results beyond expectations. The hierarchical nature of hyperbolic space aligns well with the hierarchical generation of textual semantics by the HTP method, leading to superior guidance.

Finally, integrating the MD into the model, leveraging visual–textual features during decoding, resulted in significant performance improvements of 2.05%, 2.14%, and 2.57% on the three datasets, respectively. This robustly demonstrates the effectiveness of the three methods proposed in this study.

2) Sensitivity Analysis of the SCoI Ratio: In the proposed HTP framework, the proportion of semantic categories assigned to the SCoI serves as a key hyperparameter. To evaluate the robustness of this design choice, we conduct a sensitivity analysis by varying the SCoI ratio and report the corresponding performance on the LEVIR-CD and CDD datasets. The results are presented in Fig. 7, where an SCoI ratio of 0.67 corresponds to selecting the top two-thirds of semantic categories ranked by their occurrence frequency as SCoI.

As shown in Fig. 7(a) and (b), the performance on the LEVIR-CD dataset remains relatively stable across different SCoI ratios. The default setting (SCoI ratio = 0.67) achieves the best overall performance, while neighboring configurations exhibit only minor fluctuations. In contrast, on the CDD dataset, the performance variations become more pronounced as the SCoI ratio changes. This phenomenon can be attributed to the richer and more diverse semantic categories present in CDD, which make the model more sensitive to the inclusion or exclusion of specific semantic classes. Nevertheless, the overall performance trends on both datasets remain smooth, and SCoI ratios in the range of 0.5–0.67 consistently achieve a favorable balance between semantic coverage and noise suppression.

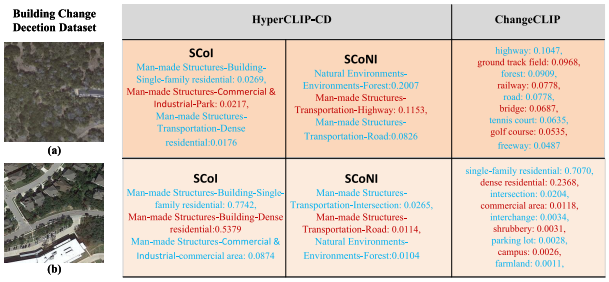


Fig. 8. Textual prompts and semantic probabilities generated by HTP and ChangeCLIP for different datasets. Both subfigures (a) and (b) are derived from the LEVIRCD building change detection dataset. In this dataset, the SCoI primarily corresponds to man-made buildings.

V. Discussion

A. Comparison of Different Text Generation Methods and Applicability Analysis

In this article, we propose an HTP generation method. Among the limited CD methods based on multimodal learning, each has its unique approach to textual generation. For instance, ChangeCLIP identifies 56 common semantics and directly generates textual prompts based on these categories, serving as the foundation for the HTP method. In contrast, [41] focuses solely on building CD tasks, constructing two text sets, "building" and "nonbuilding," to generate building-specific semantic prompts. Zhao et al. [39] took a groundbreaking approach by exploring the distinction between target changes and visual changes at the semantic level, conducting CD for a single specific semantic category. This work has been a source of inspiration for us. However, since the latter two methods are not universally applicable, this section compares the textual generation methods of HTP and ChangeCLIP.

Fig. 8 illustrates the textual prompts and semantic probabilities generated by HTP and ChangeCLIP for building change detection data. ChangeCLIP feeds the textual semantics of the top nine confidences into the text encoder as textual prompts, while HTP sends the semantics of the top three confidences in SCoI and SCoNI separately into the text encoder. As depicted in Fig. 8(a), the image comprises forests and roads, with forests occupying a significantly larger proportion than roads. This distribution is reflected in HTP, where both of these semantics are categorized as SCoNI. In the subsequent HASM process, these semantics are suppressed, as focusing on forests and roads is unnecessary for building change detection. Conversely, the semantic distribution in ChangeCLIP appears more chaotic, with a higher confidence in roads than in forests. Regarding Fig. 8(b), HTP captures the building-related semantics needed with higher confidence compared to ChangeCLIP. This is partly because ChangeCLIP introduces excessive irrelevant semantics into the network, thereby disturbing the model's decision-making process. This distinction is primarily attributed to the spatially constrained textual prompting

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TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT METHODS IN THE $F1$ INDEX ON THE LEVIR-CD DATASET

Methods	F1	Parameters(M)	FLOPs(G)
ChangeMamba-base	90.16	84.70	179.32
RSM-CD	91.10	55.81	-
ChangeCLIP(VIT/16)	91.30	86.51	66.35
LGCD	91.89	**26.14**	-
HyperCLIP-CD-v1	91.06	49.03	**38.68**
HyperCLIP-CD-v2	89.97	49.03	89.76
HyperCLIP-CD	**92.11**	49.03	45.74

TABLE VII
PERFORMANCE COMPARISON OF DIFFERENT DISTANCE METRICS WITHIN THE HMSA FRAMEWORK ON THE CDD DATASET

Methods	Embedding Space	Pre	Rec	F1	IOU
HMSA-L2	Euclidean	96.40	97.05	96.72	93.65
HMSA-Cos	Euclidean	**98.15**	96.20	97.17	94.50
HMSA-Hyp	Hyperbolic	98.05	**97.70**	**97.87**	**95.84**

strategy adopted by HTP. HTP restricts the generation of textual prompts to actual change regions, enabling the model to focus on semantically meaningful cues while effectively suppressing irrelevant background semantics. As a result, pseudo-change interference is substantially reduced, leading to more stable and discriminative multimodal alignment.

However, this reliance on change masks also implies certain limitations in few-shot or zero-shot scenarios, where accurate change localization may be difficult to obtain. In practical applications, change detection tasks are often task-driven or semantics-oriented, with the categories of interest being predefined. Under such conditions, HTP allows flexible adjustment of the semantic regions of interest according to task requirements. Moreover, when supervision is limited, unsupervised or weakly supervised change proposal strategies (e.g., SAM-based region proposals or difference-map-driven cues) can be integrated to provide approximate spatial guidance. Our primary objective is to identify the SCoI within a target dataset, and extending HTP toward more weakly supervised settings remains a promising direction for future work.

In addition, the semantics extracted by HTP share a common hierarchical root, namely "man-made structures," which further enhances semantic coherence across related categories. This hierarchical semantic partitioning mitigates CLIP's representational uncertainty in occluded or ambiguous regions, thereby providing additional robustness during multimodal semantic alignment.

B. Necessity of Hyperbolic Information on CD

Before this study, Yang et al. [48] explored the role of hyperbolic representations in RSCD tasks, demonstrating that integrating hyperbolic geometry into feature learning can enhance feature representation richness. Research in [64] further indicates that hyperbolic operations can improve the performance of remote sensing semantic segmentation, providing evidence of the effectiveness of hyperbolic priors in remote sensing image interpretation. The unique characteristics of remote sensing images, such as high nadir angles and high resolution, often lead to geometric distortions. The introduction of hyperbolic information helps compensate for features lost during the mapping of complex scenes to 2-D grids, thereby preserving critical topological information for better analysis.

In this study, we first analyze the optimal integration of hyperbolic information into the network architecture. As shown in Table VI, we compare three variants: HyperCLIP-CD-v1 (without HMSA), HyperCLIP-CD-v2 (with HMSA applied to all feature levels), and the proposed HyperCLIP-CD (with HMSA applied only to fine-grained features). All experiments in Table VI use ResNet-50 as the image backbone. The results indicate that the proposed design achieves a favorable balance among model parameters, FLOPs, and detection accuracy. In contrast, applying hyperbolic modules to all feature levels not only significantly increases computational cost but also degrades performance. This is primarily attributed to the fact that hyperbolic operations on coarse-grained features tend to introduce substantial noise, which adversely affects feature discrimination. Moreover, although hyperbolic computations do not introduce additional learnable parameters, the large spatial scale of coarse-grained features makes hyperbolic distance calculations expensive, thereby slowing down network training. These observations are consistent with the conclusions reported in [48].

Beyond architectural placement, we further investigate whether the performance gain originates from the intrinsic geometric properties of hyperbolic space rather than the HMSA framework itself. To this end, we replace the hyperbolic distance in HMSA with standard Euclidean distance (L2) and cosine similarity, while keeping all other components unchanged. To ensure a robust evaluation, the CDD dataset is employed for quantitative analysis due to its diverse semantic categories and its high representativeness of remote sensing image characteristics. The quantitative results are summarized in Table VII. It can be observed that although Euclidean and cosine-based variants achieve competitive performance, the hyperbolic version consistently delivers superior results, particularly in terms of $F1$-score and IoU. This confirms that hyperbolic distance is more effective in modeling hierarchical semantic relationships and separating change-relevant semantics from background interference in remote sensing change detection. These findings underscore the potential of hyperbolic geometry as a robust alternative for advanced remote sensing interpretation tasks.

VI. CONCLUSION

In this article, we propose a multimodal network, HyperCLIP-CD, for RSCD leveraging hyperbolic multimodal information. A specially designed textual prompting method is employed to capture the SCoI in CD tasks and generate textual prompts to guide image feature learning. The proposed HMSA module goes beyond planar space, aligning visual–linguistic features in hyperbolic space and introducing hyperbolic information to enhance the semantic coherence of multimodal features. Additionally, we design a multimodal decoding

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architecture that fully utilizes semantic information during the decoding phase and enables HyperCLIP-CD to address both BCD and SCD tasks. To evaluate the performance of HyperCLIP-CD, we conduct comprehensive experiments on two BCD datasets (CDD and LEVIR-CD) and one SCD dataset (SECOND). The experimental results demonstrate that our method outperforms state-of-the-art approaches on both tasks. Furthermore, we systematically analyze the performance of each module in HyperCLIP-CD and provide insights into their effectiveness.

Finally, we observe that multimodal approaches are becoming increasingly popular in the RSCD. However, the application of multimodal learning to RSCD still requires further development, as generating semantic descriptions from a single image provides limited semantic information. In the future, we aim to explore more detailed semantic generation to improve the generalizability of CD datasets. Hyperbolic space has long been a prominent focus in NLP, and it has recently shown impressive performance in remote sensing image interpretation tasks. The integration of multimodal learning with hyperbolic space presents significant potential for further research.

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